

Lab 4: AI Delegation Conversations

Arjun Srikanth | PES1UG24AM920 | Section H

Assigned game: SETAPESU26/08_helicopter

Primary implementer and coordinator: Codex

Subagents: Volta (test engineer), Turing (collision reviewer)

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Scope: all 16 recorded outward messages from these two AI delegation sessions. The three prompts were authored by Codex, not the student. Student messages are not included. This is not a student-to-AI transcript. Codex implemented the game; subagents contributed tests and review. Internal reasoning, tool logs and injected environment metadata are excluded. Only local paths/link targets and dash typography are normalized for presentation; message wording, authorship, chronology and reported limitations are preserved. Recorded session times are shown in IST. This delegation-only scope may not satisfy an instructor request for the complete student-to-AI chat history.

Current submission status - verification note, not a historical message

The final simplified game passes all 18 tests. It uses one-hit shield consumption and removal of the hit obstacle, not the historical contact-immunity/per-wall models discussed below. The original before video and final after video are unchanged. These recorded conversations preserve earlier stages and do not claim a new subagent review of the final simplification.

1. Test engineer - Volta

Historical test development and initial shield review; this shield model was later revised.

Coordinator (Codex) - prompt | 2026-10-07 13:11:39 IST

You are the bounded test engineer for SE Lab4 helicopter. Work in [submission checkout]. Write only Lab-4/helicopter/tests/test_game.py (use apply_patch). Do not edit game files, commit, push, install, or create user prompts. Original game source is [starter source]. Inspect that source. Planned API: Helicopter.handle_input(keys), .update(height), .get_rect(), .shield_active bool initially False. GameEngine.game_over bool, .distance float initially 0, .handle_keydown(pygame.K_r) resets after game over, .handle_keydown(pygame.K_SPACE) activates shield during play, .update() frame movement/scoring. SCROLL_SPEED remains 3 and screen 700x500. Movement will use MAX_SPEED=5, acceleration .4, immediate direction reversal (discard previous wrong-signed vy), damping on release, center-y bounded height/2..HEIGHT-height/2. Collision should include exact touching of wall edge ($\leq$ $\geq$); gap is safe when body fits strictly, helicopter/collision follows Rect integer geometry; shield protects one contiguous contact with one wall and is consumed instantly, overlap in following frames is same contact, separation/reentry or another wall kills until rearmed. Write meaningful unittest tests for speed cap, immediate reversal, both boundaries, release damping, top/bottom wall collision, both gap margins safely inside, exact-touch collision, frozen game over, restart, distance accumulation/reset, shield consumed exactly once, persistent overlap safe, reentry/second distinct wall lethal, ability rearm on a new Space. Keep tests deterministic by set frames_until_spawn large and seed/random where needed. Include setup dummy SDL env before import, src path via pathlib so 'python -m unittest discover -s tests -v' from helicopter works. Do not inflate redundant tests. Run once implementation becomes available or report tests built with any intentional baseline failures. Main agent is capturing unchanged original video now; your write scope is disjoint.

Subagent - reply | 2026-10-07 13:11:41 IST

I'll inspect the starter helicopter source and the current test layout, then write only Lab-4/helicopter/tests/test_game.py with deterministic unit tests matching the planned API. I'll run discovery if the implementation is present; otherwise I'll report any expected baseline failures.

Subagent - reply | 2026-10-07 13:12:01 IST

The starter confirms the current implementation lacks the Lab 4 behaviors, and the submission has no helicopter source yet. I'm writing the tests against the planned contract, using only `test_game.py`; collision cases will use fixed obstacles and disable spawning so frame outcomes stay deterministic.

Subagent - reply | 2026-10-07 13:12:30 IST

The test module is now in place with 15 focused cases covering movement, boundaries, collision geometry, game state, scoring, and shield lifecycle. Since the submission helicopter source is not present yet, I'm doing a syntax check and verifying that only the requested test file was changed.

Subagent - reply | 2026-10-07 13:12:37 IST

The syntax check passed. The worktree also contains the main agent's unrelated `before.mov`; I left it untouched. My syntax check generated a `__pycache__`, so I'm removing only that generated cache to keep the requested write scope clean.

Subagent - reply | 2026-10-07 13:12:42 IST

Created [test_game.py](#) with 15 deterministic tests covering all requested behaviors.

- Syntax check passed.
- Full unittest run deferred because submission game source is not yet available.
- No game files were edited.

Coordinator (Codex) - prompt | 2026-10-07 13:16:01 IST

Implementation is now complete in your same checkout. Review read-only Lab-4/helicopter/game and main.py against original README, plus test file. Full suite has 14 passing tests. Check bugs especially exact edge collision, gap safety, shield continuous contact versus repeated collision/re-entry, GUI HUD fit, restart state. Two test corrections by root: exact lower touch y337 not 363; rearm test move to other wall before update so one continuous already-absorbed contact does not consume rearmed shield. Report concrete issues; do not edit or commit. Root is recording live game while you review.

Subagent - reply | 2026-10-07 13:16:05 IST

I'll perform a read-only review of the completed Lab 4 implementation, comparing `README.md`, `game/`, `main.py`, and the tests. I'll specifically trace integer-rect edge cases, strict gap containment, shield contact identity/re-entry, restart reset coverage, and HUD layout, then report only concrete findings without modifying files.

Subagent - reply | 2026-10-07 13:16:12 IST

The code review shows the core contact model is intentionally stateful by `(obstacle, side)`: persistent overlap remains absorbed, while separation removes the absorbed contact and permits re-entry to be lethal. I'm now running the corrected 14-test suite and a small read-only render/layout probe to validate the remaining concerns.

Subagent - reply | 2026-10-07 13:16:23 IST

The requested suite cannot execute in this shell because pygame is not installed (`ModuleNotFoundError`); I will not install it per scope. Static review is otherwise complete, and I'm doing one final targeted check for overlooked state-reset or layout defects before reporting findings.

Subagent - reply | 2026-10-07 13:16:39 IST

Read-only review completed.

No concrete implementation bugs found:

- Exact wall-edge contact uses inclusive comparisons correctly.
- Gap passage requires strict body containment.
- Shield absorption persists for continuous contact, expires on separation, and re-entry/another wall is lethal.
- Restart resets game state, distance, obstacles, and shield state.
- HUD strings fit within the fixed 700×500 window.
- Corrected tests contain 14 cases.

I could not rerun the suite because this shell lacks pygame (`ModuleNotFoundError`); no packages or files were changed.

2. Collision reviewer - Turing

Historical review of the per-wall clearing revision; that implementation was later simplified.

Coordinator (Codex) - prompt | 2026-10-07 13:33:48 IST

Read-only targeted review of Lab4 collision revision. Checkout [submission checkout], Python [Python environment]. User saw apparent wall phasing in earlier video. Current changed code removes continuous-contact immunity: Space shield consumes once and `Obstacle.clear_wall(side)` makes only hit top/bottom wall empty, shows one-second HUD message. Other solid wall still lethal. HUD outside 500px playfield; `WINDOW_SIZE` 700x600. Review `game_engine`, `helicopter`, `obstacle`, `renderer`, `main` and tests against `Lab-4/STARTER_README.md`. Run suite using provided interpreter, and test several random seed simulations if useful. Look for actual solid-wall bypass, reactivation/restart, exact edges/gap/bounds, HUD problems. Do not edit files, commit/push, share threads, or produce synthetic screenshots. Report any concrete bug or explain clean result and independent tests. Root is independently recording and reviewing real gameplay; your task is code/logic verification.

Subagent - reply | 2026-10-07 13:33:50 IST

I'll perform a read-only review in the specified checkout, compare the changed logic and tests with `STARTER_README.md`, then run the provided test suite and a few deterministic simulation checks. I won't modify files or external state.

Subagent - reply | 2026-10-07 13:34:00 IST

The changed code is confined to collision handling, obstacle wall clearing, HUD layout, and tests; the checkout has no uncommitted changes beyond that revision. Initial inspection shows the intended one-wall clearing model is present, so I'm now validating edge geometry, post-hit state transitions, rendering bounds, and runtime behavior with the supplied interpreter.

Subagent - reply | 2026-10-07 13:34:14 IST

The full suite passes all 18 tests, including exact-touching collision, exhaustive gap positions, long-held boundary checks, shield rearming, post-shield lethality, restart freezing/reset, and scrolling-wall collision. Independent probes also rendered 20 seeded runs without exceptions, preserved the 500px arena bounds, and confirmed side-specific wall clearing. I'm doing a final status/diff audit now to ensure the review itself left no artifacts and to check for any remaining concrete defect.

Subagent - reply | 2026-10-07 13:34:20 IST

Read-only review complete. No concrete bugs found.

Checks performed:

- Compared changed code against Lab-4/STARTER_README.md.
- Ran all tests: 18 passed.
- Ran `git diff --check` and Python compilation successfully.
- Ran 20 deterministic seeded simulations with rendering.
- Verified:
 - Solid walls remain lethal after shield consumption.
 - Only the struck wall is cleared.
 - Shield reactivation and restart work correctly.
 - Exact wall contact is lethal.
 - The entire gap remains safe.
 - Helicopter stays within the 500px playfield.
 - HUD renders within the 700×600 window.
 - Game-over state freezes movement and distance.
 - Shield-hit HUD message persists for approximately one second.

No files were edited, committed, pushed, or shared.